



Fig. 4: Block diagram of RFID-based automatic vehicle parking system

Tags (transponders). Fig. 3 shows the internal structure of a typical RFID tag. It comprises a microchip containing identifying information about the item and an antenna that transmits this data wirelessly to the reader. At its most basic, the chip contains a serialised identifier or licence plate number that uniquely identifies that item (similar to bar codes). A key difference, however, is that RFID tags have a higher data capacity than their bar code counterparts. This increases the options for the type of information that can be encoded on the tag; it may include the manufacturer's name, batch or lot number, weight, ownership, destination and history (such as the temperature range to which an item has been exposed). In fact, an unlimited list of other types of information can be stored on RFID tags, depending on the application's requirements.

RFID tag can be placed on individual items, cases or pallets for identification purposes, as well as fixed assets such as trailers, containers and totes. There are different types of tags with varying capabilities:

1. Read-only tags contain such data as a serialised tracking number, which is pre-written onto these by the tag manufacturer or distributor. These are generally the least expensive tags as no additional information can be included when they move through the supply chain. Any update to the information has to be maintained in the application software that tracks the stock-keeping unit's movement and activity.

2. Write-once tags enable the user to write data once in the production

over the original data—when needed. Examples include the time and date of ownership transfer or updating the repair history of a fixed asset. While these are the most costly of the three tag types and impractical for tracking inexpensive items, future standards for electronic product codes (EPCs) appear to be headed in this direction.

Other features of the tag include:

Data capacity. The capacity of data storage on a tag can vary from 16 bits to several thousand bits. Of course, the greater the storage capacity, the higher the price of the tag.

Form factor. The tag and antenna structure can come in a variety of physical form factors and can either be self-contained or embedded as part of a traditional label structure (termed as 'smart label,' it has the tag inside what looks like a regular bar code label).

Passive and active. Passive tags have no battery and broadcast their data only when energised by a reader. It means these must be actively polled to send information. Active tags broadcast data using their battery power. This means their read range is greater than passive tags—around 30 metres or more, versus 5 metres or less for most passive tags.

The extra capability and read range of active tags, however, come at a cost. These are several times more expensive than passive tags. Today, active tags are much more likely to be used for high-value items or fixed assets such as trailers, where the cost is minimal compared to item value and very long read ranges are required. Most traditional supply chain applications,

or distribution process. The data may include a serial number or lot or batch number.

3. Full read-write tags allow new data to be written to the tag—even

such as the RFID-based tracking and compliance programmes emerging in the consumer goods retail chain, use the less expensive passive tags.

Frequency range. Like all wireless communications, there are a variety of frequencies or spectra through which RFID tags communicate with readers. Again, there are trade-offs among cost, performance and application requirements. For instance, low-frequency tags are cheaper than ultra-high-frequency (UHF) tags, use less power and are better able to penetrate non-metallic substances. These are ideal for scanning objects with high water content, such as fruit, at close ranges.

UHF's typically offer longer range and can transfer data faster. But these use more power and are less likely to be effective with some materials.

Electronic product code (EPC) tags. EPC is an emerging specification for RFID tags, readers and business applications. It represents a specific approach to item identification, including an emerging standard for the tags—with both the data content of the tag and open wireless communication protocols.

RF transceiver. RF transceiver is the source of RF energy used to activate and power the passive RFID tags. It may be enclosed in the same cabinet as the reader or it may be a separate piece of equipment. When provided as a separate piece of equipment, the transceiver is commonly referred to as an RF module. RF transceiver controls and modulates the radio frequencies that the antenna transmits and receives. The transceiver filters and amplifies the backscatter signal from a passive RFID tag.

How this vehicle parking system works

Fig. 4 shows the block diagram of the RFID-based automatic vehicle parking system.

To get started with RFID-based automatic vehicle parking system, the vehicle owner has to first register the vehicle with the parking owner and get the RFID tag. When the car